



DPUMiniAcc

Simple ACC system (Adaptive Cruise Control).

This ACC system allows simple longitudinal control of a driver selected target velocity. In addition, the system maintains a safe distance to a leading vehicle.

1. Inputs from DPUAccSGSensor or DPUTRFDataSmall:

<i>Nearest</i>	<i>true</i> , if a leading vehicle is present, else <i>false</i>
<i>NearestD</i>	Distance to the leading vehicle along the driver's lane (m)
<i>NearestV</i>	Velocity of the leading vehicle (m/s)
<i>SensorRange</i>	Maximum sensor range. The ACC system reacts on leading vehicles if <i>Nearest</i> AND (<i>NearestD</i> <= <i>SensorRange</i>)

2. Inputs from the vehicle dynamics interface VDyn:

<i>v</i>	Current velocity of the EGO-vehicle (m/s)
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3. Inputs from the Mockup interface:

<i>GasPedal</i>	Position of the accelerator pedal
<i>Brake</i>	State of the brake pedal
<i>Set</i>	If this input changes from false to true, the ACC is activated. The target velocity is set to the current velocity of the EGO-vehicle. The ACC system is deactivated by pressing the brake pedal.

4. Dynamic parameters:

<i>kv1, kv2, kd1, kd2</i> <i>kxv1kmh, kxv2kmh</i>	Velocity dependent controller parameters for DPUDVControl (see functional principle).
<i>axTargetMax[1..3]</i> <i>axv[1..3]kmh</i> <i>axTargetMin[1..2]</i> <i>axTargetTMMin</i> <i>axTargetTMMax</i>	Velocity dependent upper limits for accelerations. generated by DPUAxControlLUT.
<i>BiasGasPedal</i>	The ACC allows the driver to override if <i>GasPedal</i> > <i>BiasGasPedal</i> . Default: 0.1
<i>BiasBrake</i>	The ACC is deactivated if <i>BrakePedal</i> > <i>BiasBrakePedal</i> . Default: 0.01
<i>vMaxkmh</i>	Maximum velocity up to which the ACC works (km/h). De-fault: 180 km/h
<i>vMinkmh</i>	Minimum velocity for the ACC system (km/h). Default: 30 km/h



TDistTarget Desired time distance to the leading vehicle (s). The distance chosen by the ACC will be $d_{min} + v * TDistTarget$. Default: 2.0 s

5. Outputs for DPUDVControl:

ActivateDVControl *true*: DPUDVControl is activated; *false* otherwise
TDistTargetOut Current time distance (s)
vTarget Current target velocity. Will be set to the EGO-vehicles velocity when the ACC is activated (m/s).
vTO Velocity of the leading vehicle (m/s)
dTO Distance of the leading vehicle (m)
dMin Safety distance (m). The distance that will be chosen by the ACC system is $d_{min} + v * TDistTarget$. Default: 10 m
kv Controller parameter.
kd Controller parameter.

6. Outputs for DPUAxControlLUT:

axTargetMin Upper limit for decelerations (velocity dependent) (m/s²)
axTargetMax Upper limit for accelerations (velocity dependent) (m/s²)
axTargetDotMin Upper limit for slope of decelerations. (m/s³)
axTargetDotMax Upper limit for slope of accelerations. (m/s³)

7. Outputs:

State Current state of the ACC system:
0 Idle
1 ACC is active
2 Driver overrides the system by pressing the accelerator pedal
FU Driver presses the accelerator pedal
FB Driver presses the brake pedal

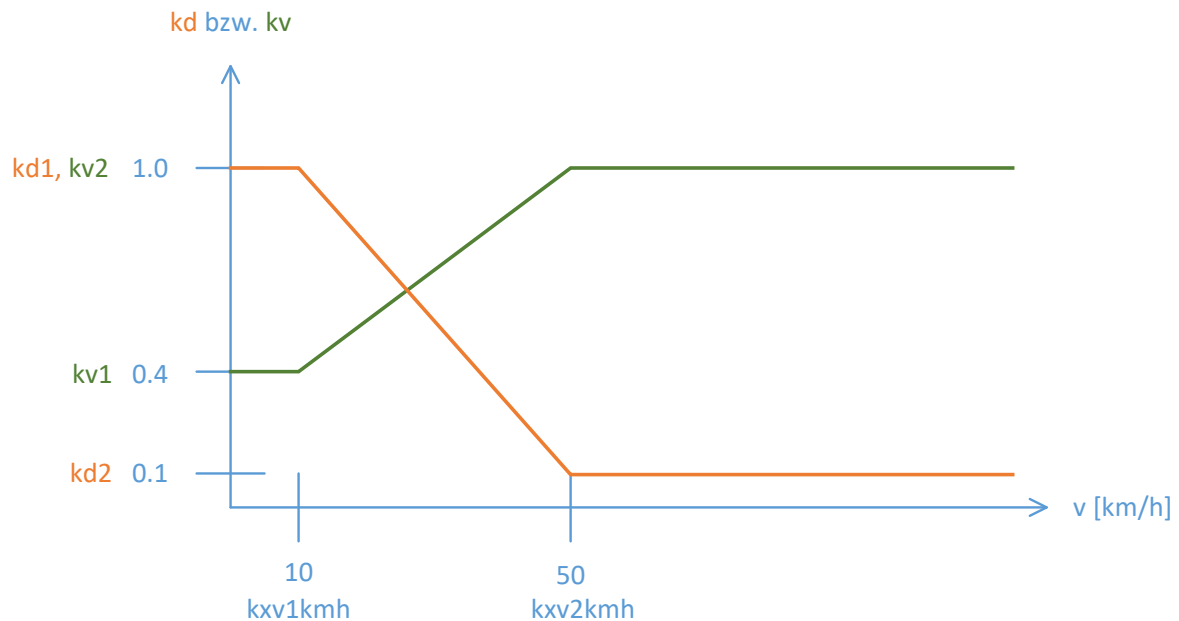
8. Functional principle:

- The ACC is activated by setting *Set* = *true*. The current velocity of the EGO-vehicle will be taken as the target velocity *vTarget*.
- Leading vehicles are recognized up to a distance of *SensorRange* along the lane of the EGO-vehicle.
- If there is a leading vehicle which is slower than the EGO-vehicle, the distance to this vehicle is controlled by the system. Thereby, the distance consists of a minimum safety distance *dMin* and a velocity dependent distance based on *TDistTarget*.
- If there is no leading vehicle within the sensor range the system controls the velocity to be *vTarget*.
- The driver can override the system by pressing the accelerator pedal.
- The driver deactivates the system by pressing the brake pedal.



9. Dependencies:

- Information about the leading vehicle are delivered by *DPUACCSGSensor* or *DPUTRFDataSmall*.
- The distance/velocity control is performed by *DPUDVControl*. The parameters *kv* and *kd* for *DPUDVControl* are adapted depending on the current velocity as shown in the following figure (the curves are based on the default values for *kd1*, *kd2*, *kv1*, *kv2* und *kxv1kmh*, *kxv2kmh*):



- The acceleration/deceleration control is performed by *DPUAxControlLUT*. Please note that this DPU is based on the vehicle dynamics simulation *DPUmstVeda*

The following figure shows the dependencies and DPU connections of an ACC system:

